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Lab Report: Lab 01 The Price of One Request

1. Prediction and Measured Token Values (Part 1)

- **Prediction:** It was predicted that the English text would be the shortest, Russian would take an intermediate position, and Kazakh would require the highest number of tokens due to its complex agglutinative structure and specific characters (ә, Ғ, Қ, Һ, Ө, Ұ, Ү, Һ, і).
- **Measured Values (Input Tokens):**
 - English (EN): **100** tokens
 - Russian (RU): **130** tokens ($1.30 \times$ of EN)
 - Kazakh (KK): **236** tokens ($2.36 \times$ of EN)

2. Annual Costs at Estimated Volume

- **Volume Justification:** A volume of 2,000 requests per day is chosen as a realistic daily load metric for a mid-sized retail bank customer support center.

Model	English (USD/year)	Russian (USD/year)	Kazakh (USD/year)
Haiku-4.5	\$1,168	\$1,190	\$1,267
Sonnet-5	\$2,336	\$2,380	\$2,535
Opus-5	\$5,840	\$5,949	\$6,336
Fable-5.1	\$11,680	\$11,899	\$12,673

Note on methodology: Input tokens were genuinely measured via the API (model.count_tokens()). Output tokens could not be dynamically measured due to project-level API access errors (403/404) encountered during generation attempts (generate_content). Consequently, a standardized assumption of 300 output tokens across all languages was applied for cost projection purposes. Therefore, the total cost figures and ratios should be treated as an approximation, whereas the reliable empirical finding of this lab is the input token growth ratio ($1.00\times / 1.30\times / 2.36\times$).

3. Model Selection for the Kazakh-Language Support Queue

For production in the Kazakh-language support queue, it is optimal to deploy the **Sonnet-5** model (or **Haiku-4.5** under strict budget constraints): it provides an optimal balance between manageable costs for processing increased token counts and high quality in following complex system prompt instructions (minimizing the risk of hallucinations regarding dates and deposit rates).

4. Cost-Reduction Lever

This lab session did not utilize **Prompt Caching**, which significantly reduces expenses for repeatedly sending invariant system prompts across high volumes of incoming requests.

AI-Use Declaration

- **Tools Used:** Personal AI assistant based on Gemini models.
- **Purposes of Use:**
 1. Adapting and verifying token-counting scripts via API in Google Colab, as well as handling model version updates.
 2. Automating annual cost calculations based on the acquired token metrics and standard pricing tiers.
 3. Structuring and formatting the lab report according to the course repository requirements.